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Tzu-Hsien Lee

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Research Interests

Robot Learning for Real-World Manipulation; Data-Efficient Imitation Learning; Vision-Based Perception and Action; Learning-Based Robot Positioning

Education

M.S. in Robotics, University of Minnesota, Twin Cities 09/2023 - Present

- Advisor: Prof. Karthik Desingh
- Courses: Robot Vision, Computer Vision, Robotics, Deep Learning for Robotics, Optimal Estimation

B.S. in Mechanical Engineering, National Yang Ming Chiao Tung University 09/2018 - 06/2022

- Courses: Robotics, Object Oriented Programming, Microcomputer System and Lab, Nonlinear Control

Publications

- Tzu-Hsien Lee, Fidan Mahmudova, Karthik Desingh.
  "Learning Category-level Last-meter Navigation from RGB Demonstrations of a Single Instance."
  arXiv: 2512.11173, 2025.

Presentations

- **Poster Presentation**. "Learning Object-Centric Local Navigation from RGB Demonstrations." Midwest Robotics Workshop (MWRW), TTIC, Chicago, IL, June 2025.

- **Student Lecture**. "RGB-D Networks." CSCI 5980: Deep Learning for Robot Manipulation, University of Minnesota, Minneapolis, MN, November 2024.

Experience

Graduate Researcher, Robotics: Perception and Manipulation Lab 09/2024 - Present

- Achieved category-level last-meter navigation for mobile manipulation by leading the design and implementation of a goal-conditioned policy and real-world data collection pipeline using multi-view RGB observations on Boston Dynamics Spot, resulting in an RA-L submission.
- Exploring one-shot generalization for eye-in-hand manipulation by studying in-context imitation learning from a single demonstration using wrist-mounted visual observations.
- Enabled spatially-aware navigation and downstream manipulation by building semantic topological mapping pipelines using vision foundation models on Boston Dynamics Spot.
- Accelerated real-world robot learning research by designing and implementing SpotStack, a modular software library for perception integration, data collection, and policy deployment on Boston Dynamics Spot.

Robotics Course Developer, CSCI 5551: Robotics, University of Minnesota 01/2026 - Present

- Designed and engineered a 10-stage real-robot manipulation curriculum by integrating a 6-DoF Lite6 robotic arm and a ZED 2i stereo camera for perception-to-manipulation learning.
- Enabled rapid development of vision-based manipulation pipelines by building reusable software infrastructure, including AprilTag-based camera-to-robot calibration, grasp primitives, and camera wrapper APIs.
- Demonstrated reference solutions by implementing zero-shot target selection and 6D object pose estimation using SAM-based segmentation and oriented bounding box fitting on point clouds.
- Established progressive checkpoints from single-object grasping to multi-object sequential stacking with increasing perception difficulty.
- Improved usability and reproducibility by authoring comprehensive 20-page technical documentation and starter templates.

Teaching Assistant, CSCI 5561: Computer Vision, University of Minnesota 09/2025 - 01/2026

- Designed and maintained an automated grading system for coding assignments, enabling scalable and consistent evaluation of student submissions.
- Held weekly office hours and provided technical guidance on computer vision algorithms and deep learning methods, including feature extraction, image registration, scene recognition, and CNN implementation.
- Contributed to midterm exam design and grading, and managed overall course grading logistics.

Robotics Intern, Industrial Technology Research Institute 11/2022 - 02/2023

- Established an autonomous grinding production line for irregularly shaped water faucets by designing force-adaptive trajectories using a robot arm and a force feedback sensor.
- Improved on-site calibration efficiency by designing a coordinate transformation algorithm for adaptive trajectory tuning.

Software Lead, NYCU Autonomous Underwater Vehicle (AUV) Team 02/2021 - 09/2022

- Spearheaded the design and implementation of the software architecture, control system, and navigation.
- Established a modular high-level/low-level robotics system by using an embedded system (STM32) for low-level attitude and motion control, and deploying ROS on Raspberry Pi and Jetson Nano for task planning and navigation.
- Reduced high-frequency noise in attitude estimation by 10% through implementing a gradient descent-based Madgwick filter on embedded hardware.
- Developed a real-time obstacle-aware navigation system by employing A* algorithm and sonar-based localization.

Selected Projects

One-Shot Imitation Learning via Visual Servoing 11/2024 – 12/2024

- Achieved one-shot manipulation generalization by implementing a two-stage imitation learning framework combining object-centric segmentation and learned visual servoing using PyTorch and ROS.
- Enabled closed-loop end-effector control from visual observations by training a Siamese CNN servoing policy on features extracted from a FiLM-conditioned U-Net segmentation model.
- Demonstrated generalization to unseen objects by evaluating the policy in ROS Gazebo using a Kinova robotic arm.

Visual SLAM in Dynamic Environments 04/2024 - 05/2024

- Achieved 0.005 rad rotation and 0.36 m translation error by developing a visual SLAM pipeline using ORB feature matching, FLANN search, and RANSAC-based pose refinement.
- Improved robustness in dynamic scenes by filtering moving objects using scene flow and KL divergence before pose estimation.

LiDAR-guided Autonomous Mobile Robot in Unstructured Environments 04/2024 - 05/2024

- Achieved autonomous exploration and package delivery by developing a navigation system using LiDAR-based feature extraction for perception and the Bug algorithm for path planning on a real-world TurtleBot.

Object Assembly Using Visual Servoing 11/2023 - 12/2023

- Achieved accurate object assembly by implementing a relative pose registration pipeline using visual features and the Iterative Closest Point (ICP) algorithm in CoppeliaSim with two camera-equipped robotic manipulators.

Technical Skills

Languages: C, C++, Python, MATLAB, JavaScript
Software & Tools: PyTorch, ROS, Gazebo, OpenCV, Qt, Git, Docker, SolidWorks
Robotics Platforms & Systems: Boston Dynamics Spot, FANUC robots, STM32, Raspberry Pi, Jetson Nano

Leadership & Service

Appointed Treasurer, NYCU Student Association 01/2021 - 09/2021

- Allocated and managed $30,000 budget, approved and recorded all financial transactions, and provided monthly financial reports to the Student Council.